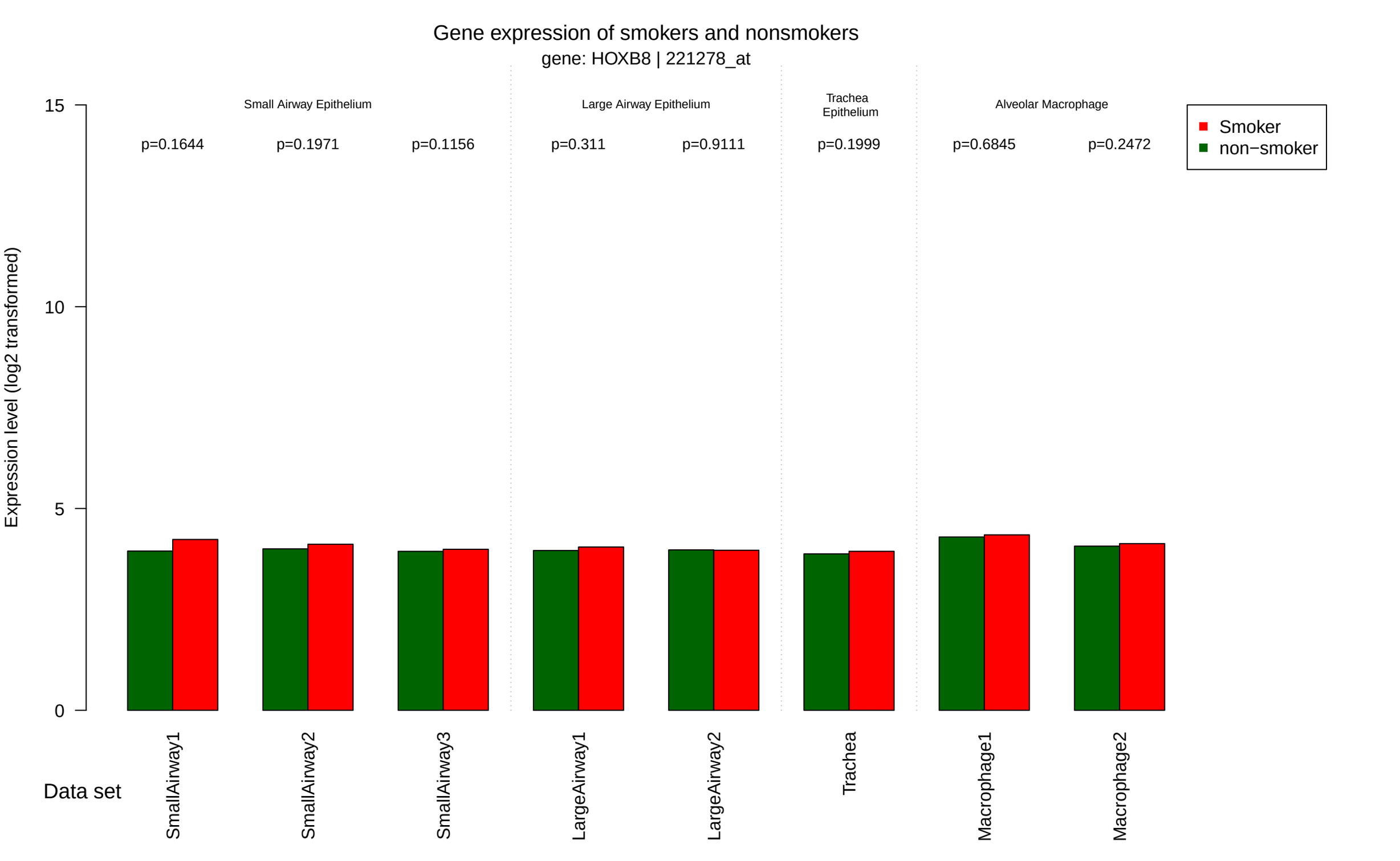
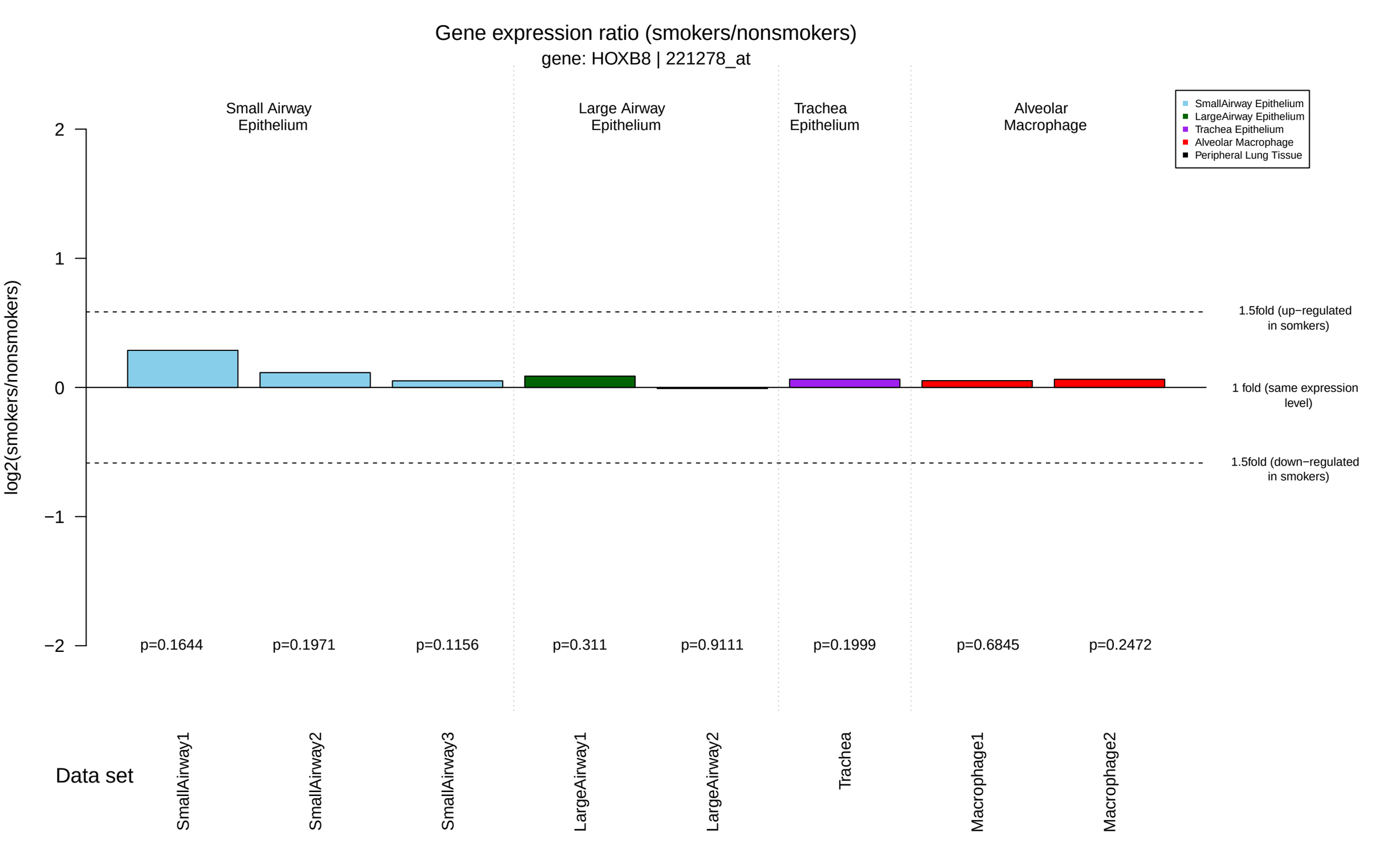
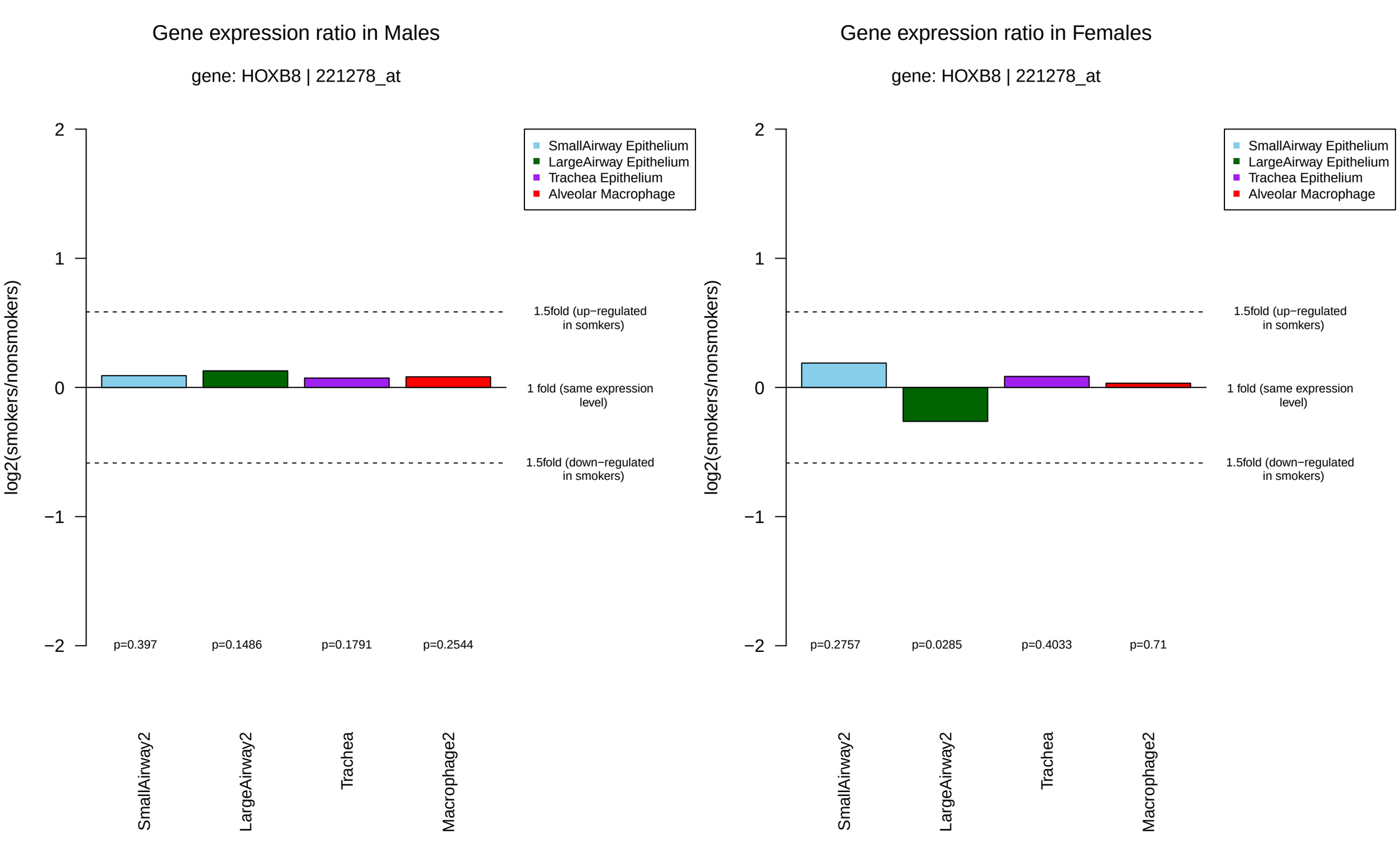
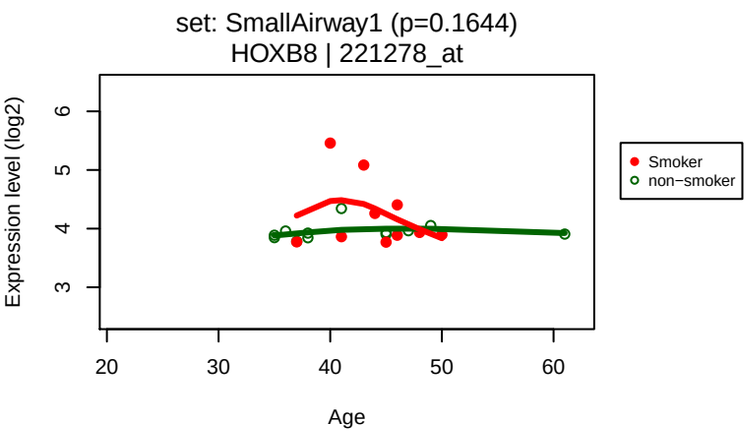
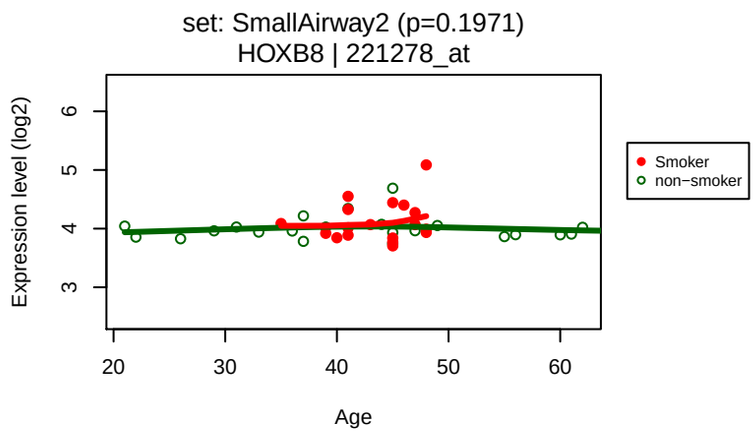
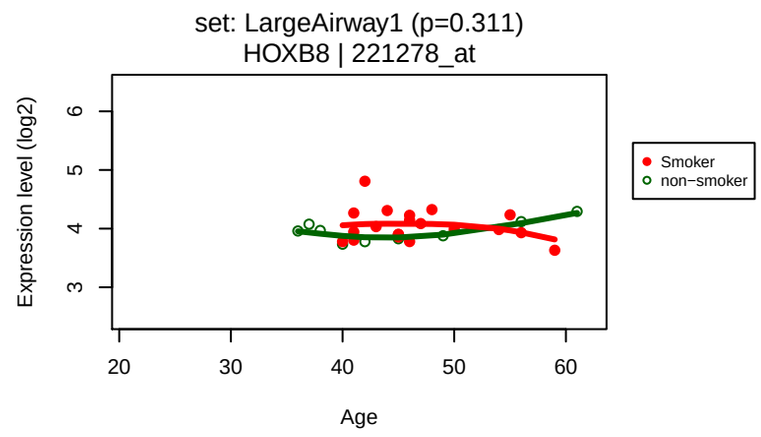

Gene expression of smokers and nonsmokers
gene: HOXB8 | 221278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1644
p=0.1971
p=0.1156
p=0.311
p=0.9111
p=0.1999
p=0.6845
p=0.2472
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

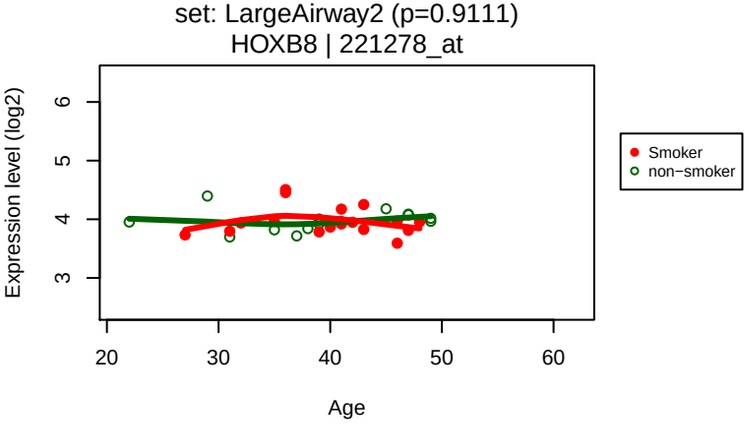

Gene expression ratio (smokers/nonsmokers)
gene: HOXB8 | 221278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.1971
p=0.1156
p=0.311
p=0.9111
p=0.1999
p=0.6845
p=0.2472
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

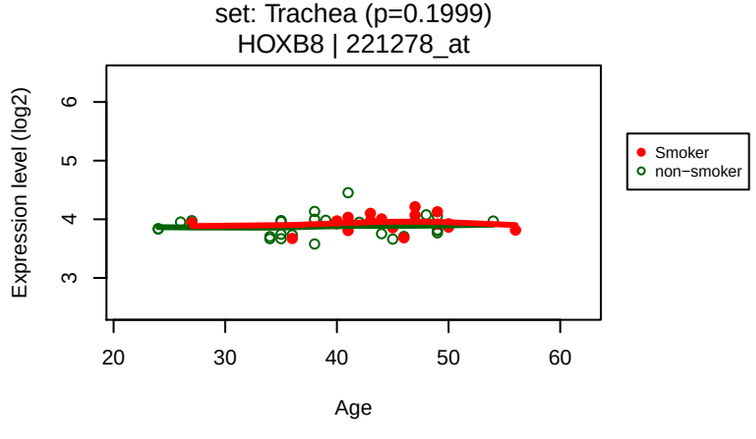

Gene expression ratio in Males
gene: HOXB8 | 221278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.397
p=0.1486
p=0.1791
p=0.2544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB8 | 221278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2757
p=0.0285
p=0.4033
p=0.71
SmallAirway2
LargeAirway2
Trachea
Macrophage2

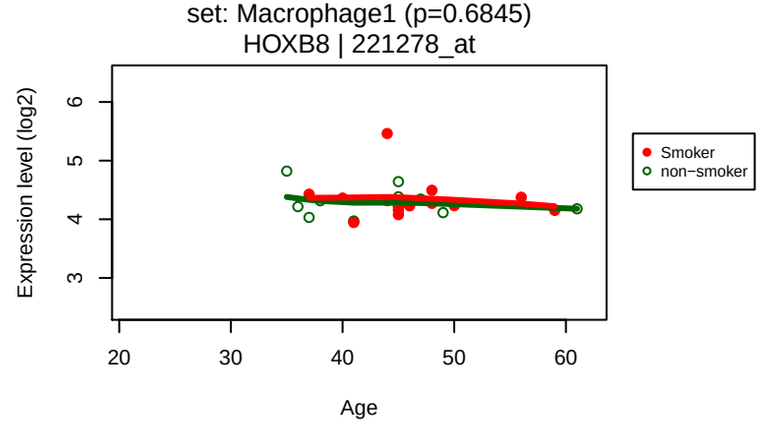

set: SmallAirway1 (p=0.1644)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker

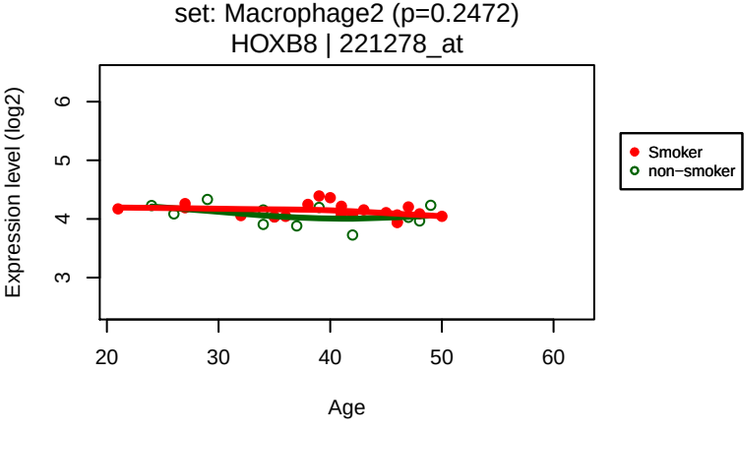

set: SmallAirway2 (p=0.1971)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker

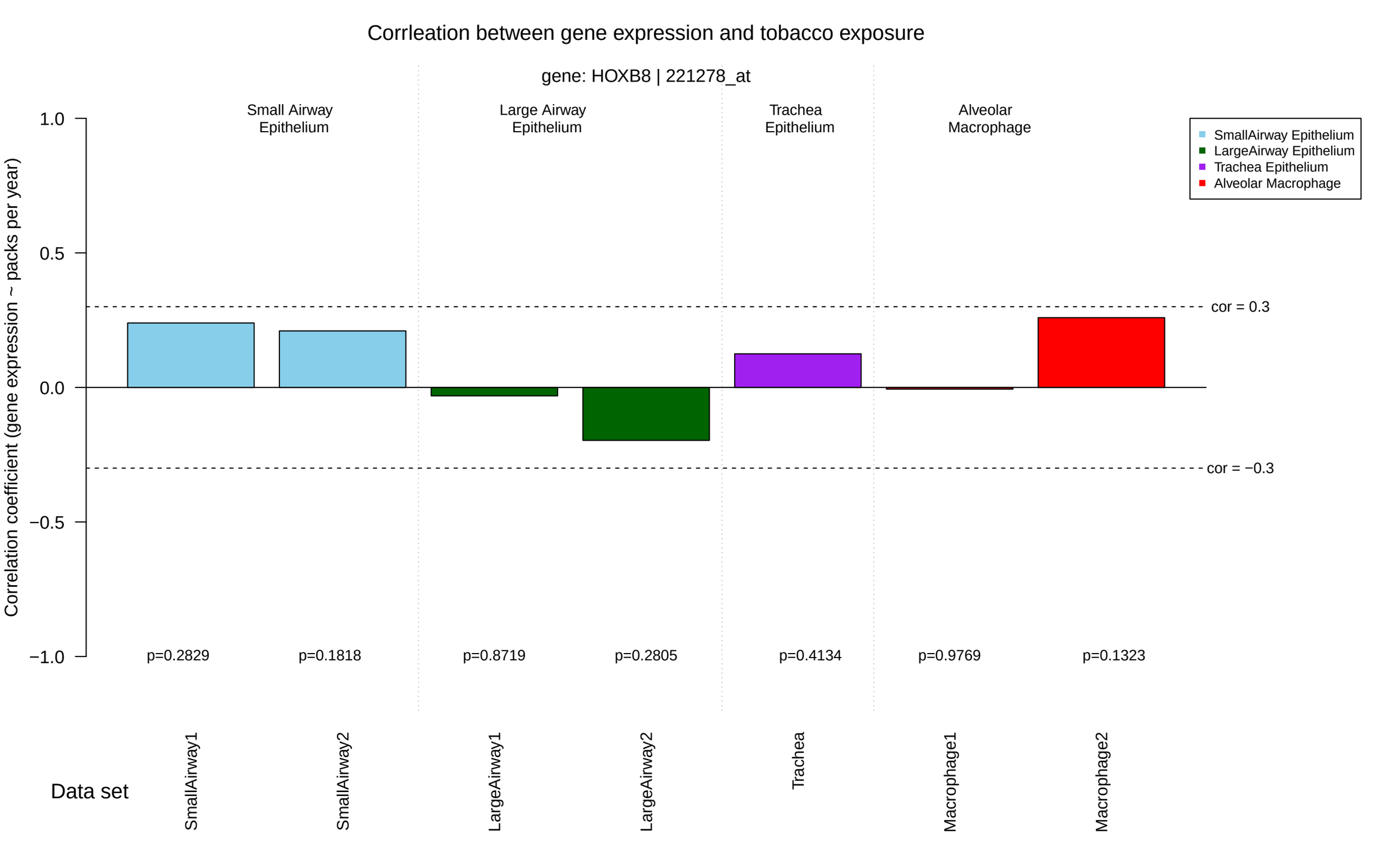

set: LargeAirway1 (p=0.311)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9111)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1999)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6845)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2472)
HOXB8 | 221278_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB8 | 221278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2829
p=0.1818
p=0.8719
p=0.2805
p=0.4134
p=0.9769
p=0.1323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2